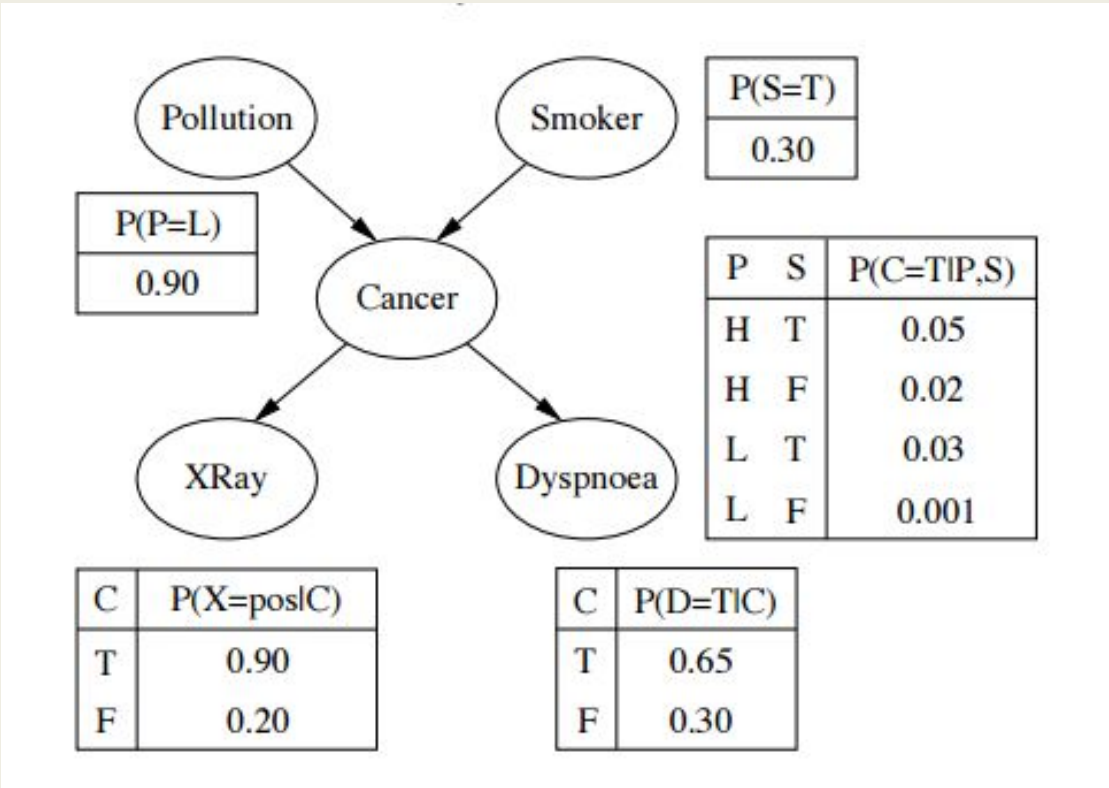
02-620
WEEK9 RECITATION

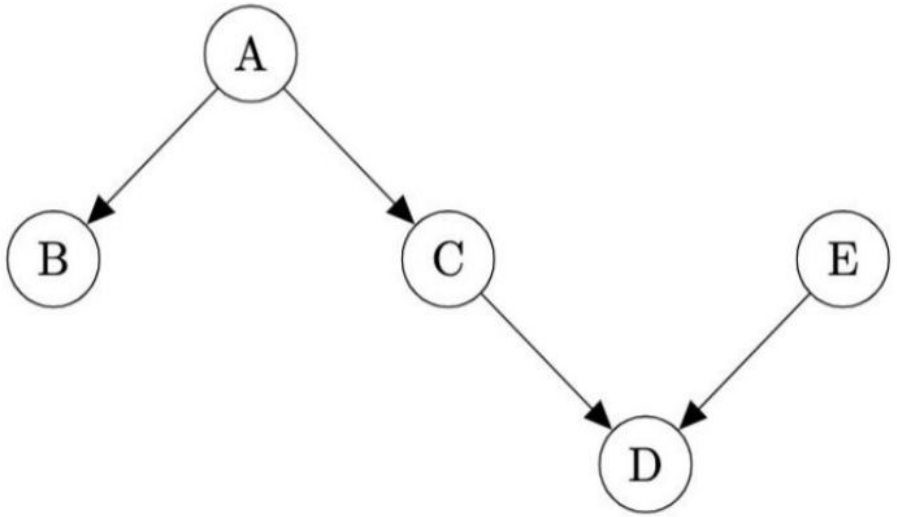
- Ruiqi Lyu

Probabilistic Graphical Models

- *Probabilistic graphical model that represents a set of variables and their conditional dependencies.*
  - *Nodes: Random Variables*
  - *Edges: Probabilistic dependencies*
- *DAGs (Directed Acyclic Graphs)*
- *Mathematical representation:*
  - $P(X_1, \dots, X_n) = \prod_i P(X_i | Pa(X_i))$
  - $Pa(X_i)$*: immediate parents of* X_i *in the graph.*



Directed Graph - factoring



$$P(X_1 \ldots X_n) = \prod_{i=1}^{n} P(X_i \mid parents(X_i))$$

$$P(A)\, P(B|A)\, P(C|A)\, P(E)\, P(D|C,E)$$

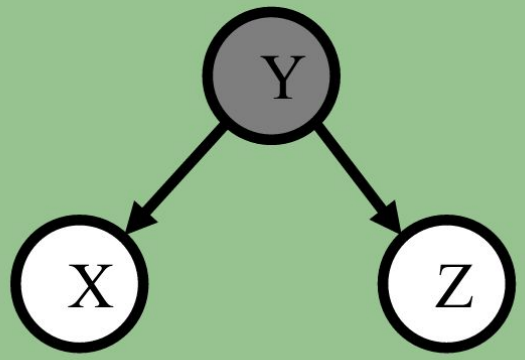
D-separation

Cascade (chain)



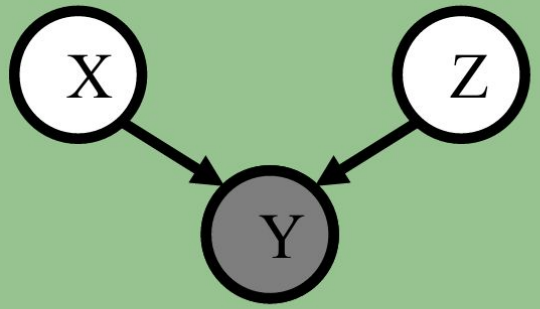
$X \perp\!\!\!\perp Z \mid Y$

Common Parent (fork)



$X \perp\!\!\!\perp Z \mid Y$

V-Structure (collider)



$X \not\perp\!\!\!\perp Z \mid Y$

Knowing Y
decouples X and Z

Knowing Y
couples X and Z

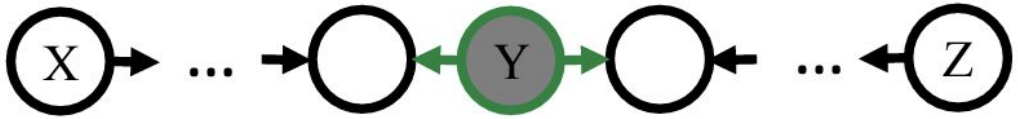
If variables X and Z are **d-separated** given a **set** of variables E

Then X and Z are **conditionally independent** given the **set** E

Definition: Variables X and Z are **d-separated** given a **set** of evidence vars E iff every (undirected) path from X to Z is blocked.

A path is blocked whenever:

1. ∃Y on path s.t. Y ∈ E and Y is a "common parent"



2. ∃Y on path s.t. Y ∈ E and Y is in a "cascade"

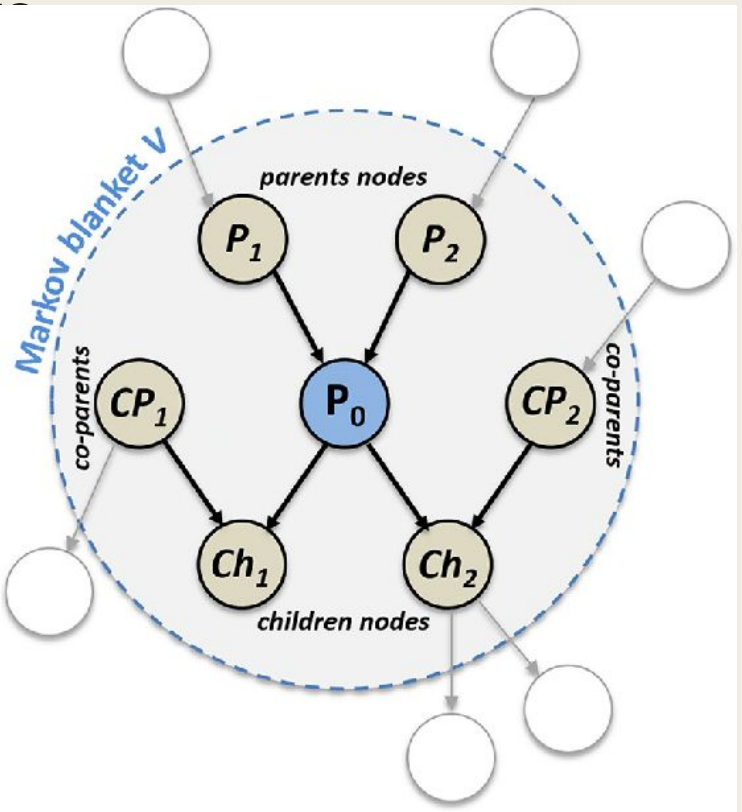


3. ∃Y on path s.t. {Y, descendants(Y)} ∉ E and Y is in a "v-structure"

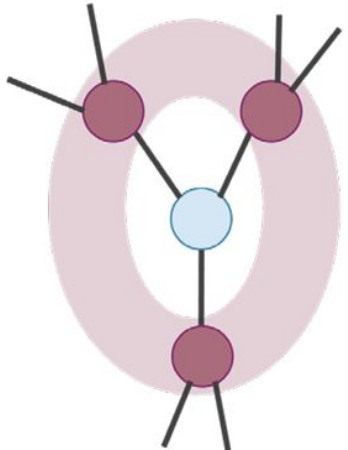


Markov Blanket

The Markov Blanket of a node in a directed graphical model is the set containing the node's parents, children, and co-parents



Undirected graphical model:



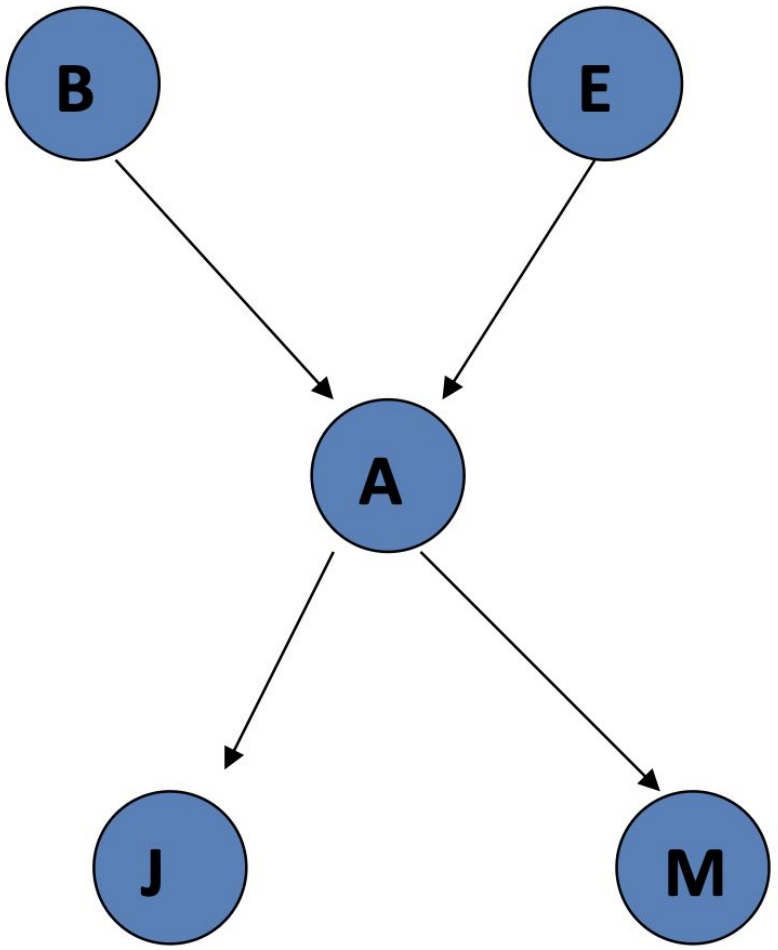
The neighbors of v are the (minimal) **Markov Blanket** of v: the smallest set of nodes S, s.t. v is conditionally independent of all other nodes, given S.

Inference

Graph Structure Inference – Structure is given, inference conditional independences

- Markov Blanket:
  - *1 vs. Multiple*
  - *Conditional independence between 1 variable and the rest of the variables*
- D-separation:
  - *1 vs. 1 (Given other variable(s))*
  - *Conditional independence between two variables*

Conditional Independence



- Markov blanket: conditional independence between one variable and the **rest of the variables**

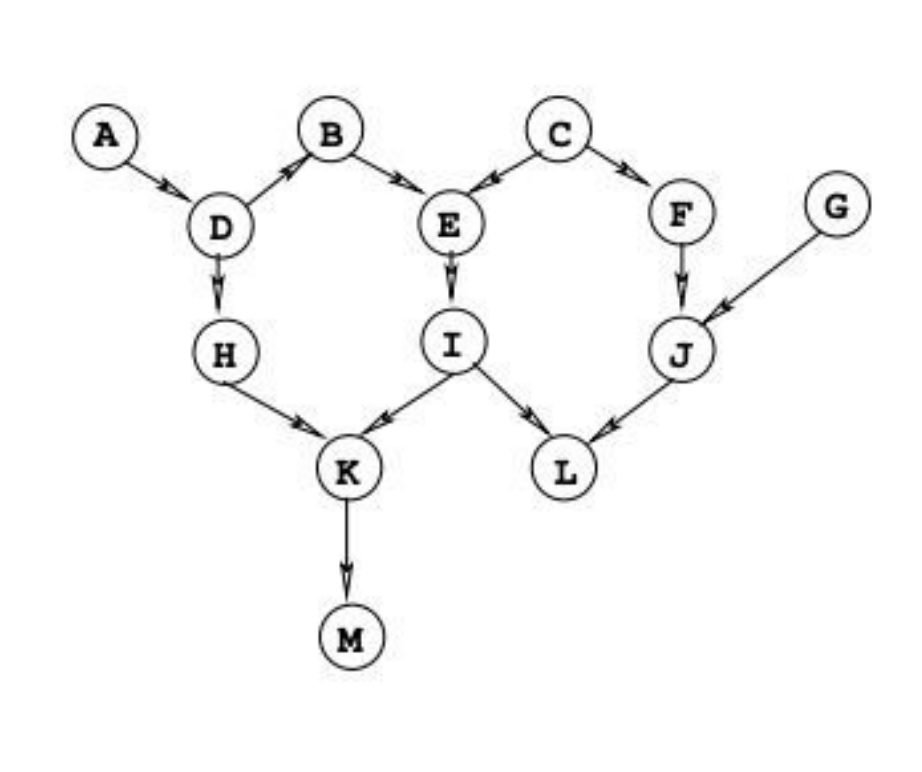
$$B \perp J, M \mid A, E$$

- D-separation: conditional independence between **two variables**

$$B \perp J \mid A$$

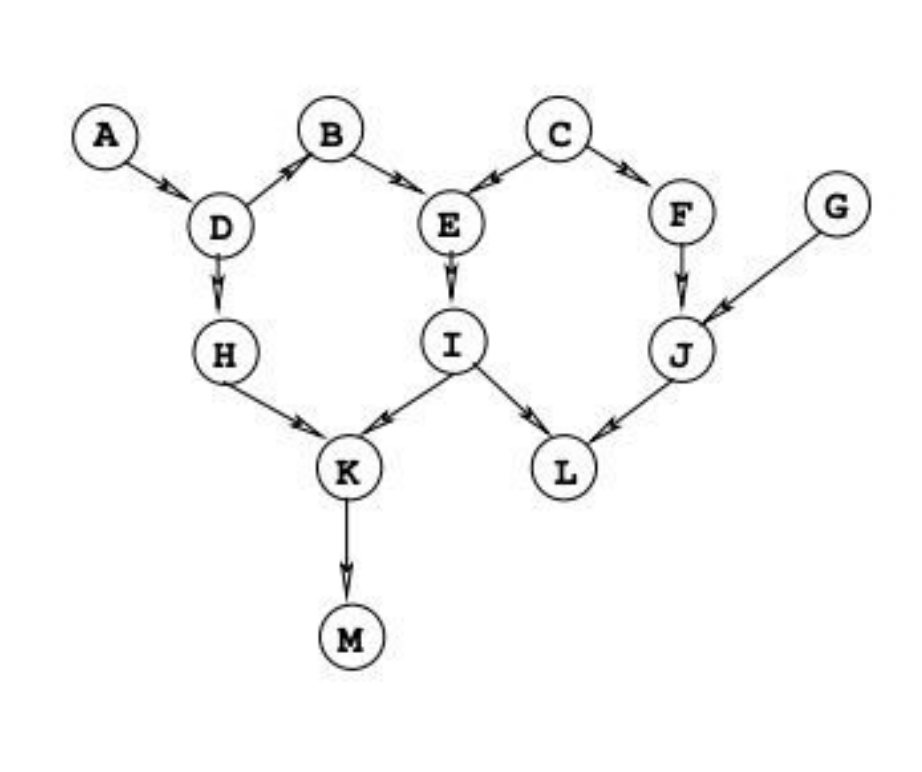
Inference

- *Is A d-separated from G given an empty set?*
- *Is A d-separated from L given M?*
- *Identify a set of nodes Z such that A is d-separated from M given Z.*
- *Is B d-separated from L given an empty set?*
- *Identify a set of nodes Z such that B is d-separated from L given Z*



Inference

- *Is A d-separated from G given an empty set? YES*
- *Is A d-separated from L given M? NO*
- *Identify a set of nodes Z such that A is d-separated from M given Z.*
- *Is B d-separated from L given an empty set? NO*
- *Identify a set of nodes Z such that B is d-separated from L given Z.*



Review: Multivariate Normal Distribution

- Recall multivariate normal distribution

$$p(x_1, \ldots, x_8) = N(\mu, \Sigma)$$

where mean vector μ and covariance matrix Σ is

$$\mu = \begin{matrix} \mu_1 \\ \vdots \\ \mu_8 \end{matrix} \qquad \Sigma = \begin{matrix} \sigma_1^2 & \cdots & \sigma_{18} \\ \vdots & \ddots & \vdots \\ \sigma_{18} & \cdots & \sigma_8^2 \end{matrix}$$

- Then, the inverse of Σ is called a precision matrix

$$\Sigma^{-1} = \Theta$$

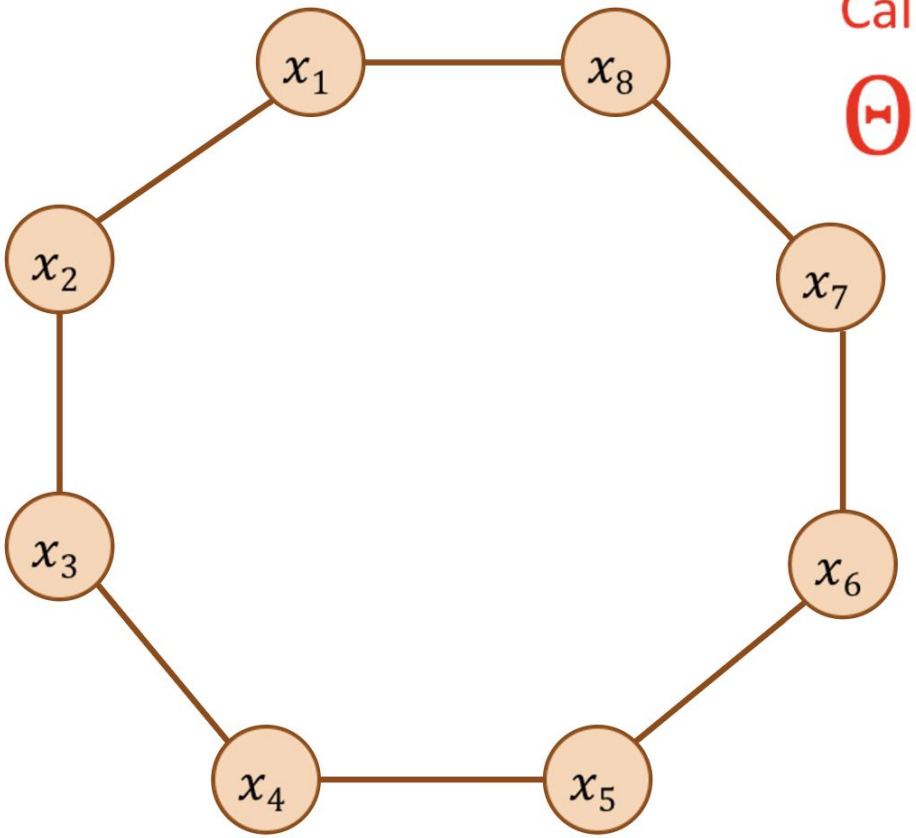
where

$$\Sigma\Theta = I$$

I is an identity matrix with
1's in diagonals
0's in off-diagonal

Example: Gaussian Graphical Models

- Gaussian graphical model



Call this

$\Theta =$

Edge weighted adjacency matrix

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
x_1	1	0.9	0	0	0	0	0	0.3
x_2	0.9	2	1	0	0	0	0	0
x_3	0	1	1.5	0.8	0	0	0	0
x_4	0	0	0.8	1.1	0.9	0	0	0
x_5	0	0	0	0.9	1.3	0.7	0	0
x_6	0	0	0	0	0.7	1.1	0.9	0
x_7	0	0	0	0	0	0.9	0.9	1
x_8	0.3	0	0	0	0	0	1	1

$$p(x_1, \ldots, x_8) = N(\mu, \Theta^{-1})$$